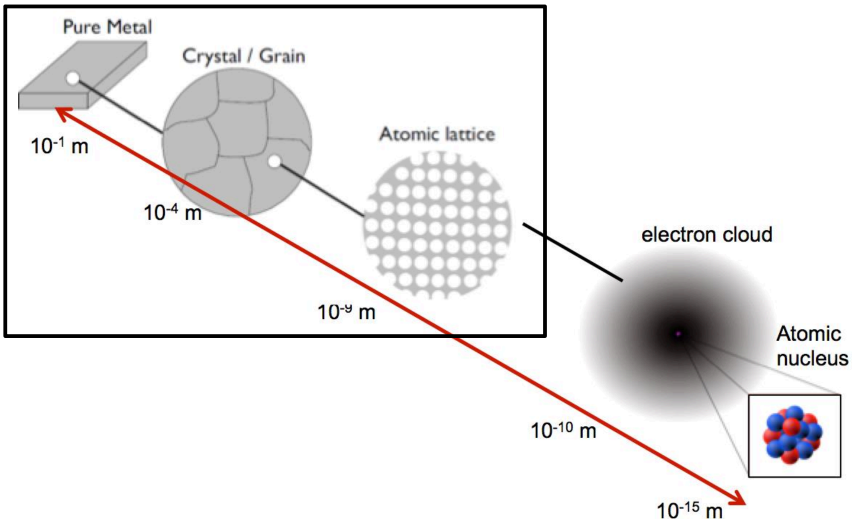
1. Basics: structure of metals

1. Structure sensitive and structure insensitive properties

There are different material properties. Some of the properties are only decided by the atom bonds (e.g. metallic bonds, ionic bonds) and crystal packing (e.g. HCP, BCC and FCC). Those structure insensitive properties are e.g. Young's Modulus, density. Some properties such as yield strength, tensile strength and ductility are also influenced by the material microstructure (e.g. grain size) caused by process parameter etc. so they are structure sensitive properties.

Mark in the figure "Hierarchy of structure" below at which scale level(s), the material properties can be changed by changing the material structures?



2. Some definitions

Polymorphism: pure metal that have different crystal structures at different temperatures, such as Ti and steel.

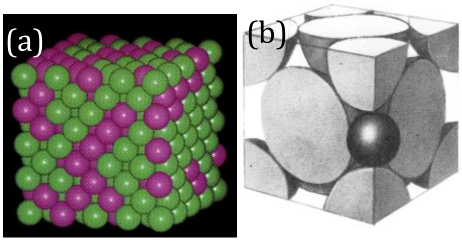
Alloys: mixture of the elements

Phase: a region with same physical and chemical properties

Phase boundary: boundary between two phases

3. Alloy: solid solutions

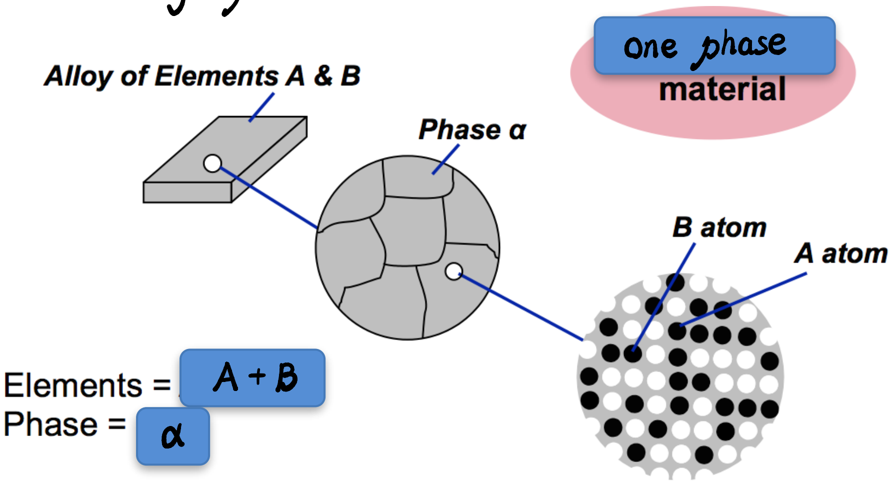
A mixture of elements can form solid solutions which are stronger than pure metals.



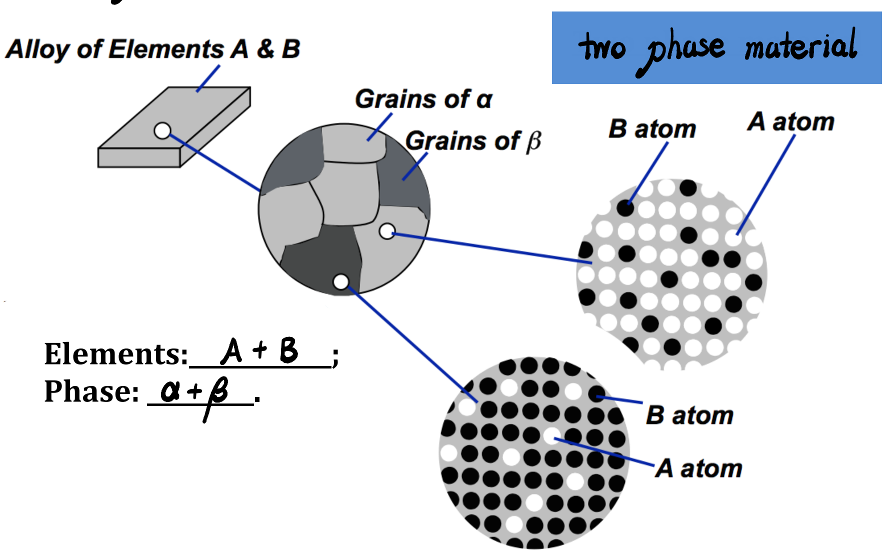
There are different types of solid solutions: ___substitutional___ solid solution (Figure a) and ___interstitial___ solid solution (Figure b).

4. Solubility limit and phases

In some cases, there is ___no___ solubility limit between two elements such as Ni and Cu, so those two elements can be mixed together ___unlimitly___ and form a ___single phase___ material.



But in many cases, there is a ___solubility limit___ and the solute precipitates out as another solution to form a ___second phase material___.



Elements: ___A + B___;
Phase: ___α + β___.

5. Grain boundary and phase boundary

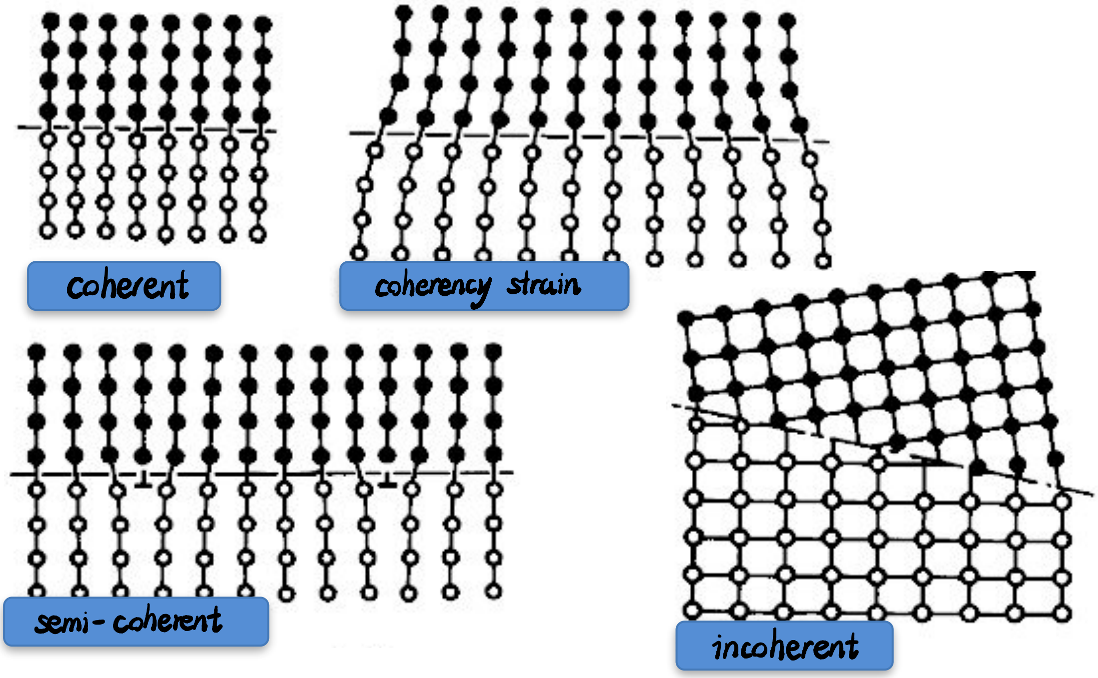
Most single-phase materials are made up of numerous ___grains___. Each grain is a region with the same ___crystal orientation___. Grain Boundaries have an ___energy___ associated with them.

Two-phase materials are also made up of grains: grains of α and grains of β.

___Each___ grain is a region with the same crystal orientation.

___Grain boundaries___ are the boundaries between two different ___grains___ and also have an energy associated with them.

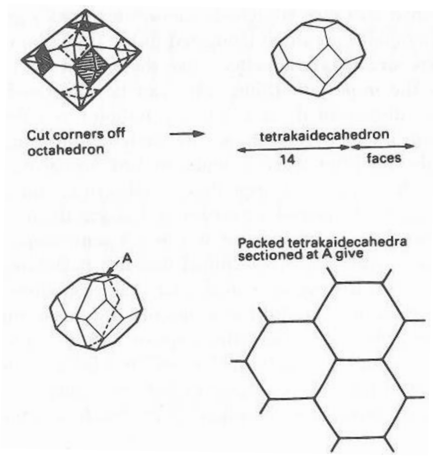
Depending on the energy associated with the two phases, there are different boundary type in metals: ___coherent___, ___coherency strain___, ___semi-coherent___ and ___incoherent___; in which ___coherent___ has the lowest energy while ___incoherent___ has the highest energy.



6. Shape of grains associated with boundary energy.

The shape of grains is usually determined by the shape that ___minimise___ the total ___interfacial energy___. E= γ_{gb}.A (γ_{gb} is grain boundary energy, A is interfacial area). When γ_{gb} is isotropic (does not depend on grain boundary orientation angle), then the lowest interfacial energy is that with the lowest interfacial area.

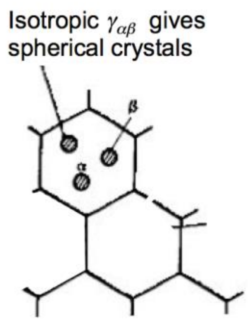
In a single phase material, grain tends to have a tetrakaidecahedron shape to ___minimise___ the interfacial area.



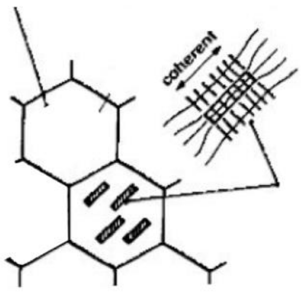
For two-phase material

- Consider phase β forming within a grain of phase α:

--- If $\gamma_{\alpha\beta}$ is isotropic, $\gamma_{\alpha\beta}$ does not depend on α or β orientation. The lowest energy interface occurs for the shape with the lowest area, leading to a sphere of phase β in phase α.



--- If $\gamma_{\alpha\beta}$ is non-isotropic, $\gamma_{\alpha\beta}$ depends on orientation between α and β. The lowest energy interface occurs for the shape with the lowest total E=$\gamma_{\alpha\beta}$A. If one interface has a lower energy, phase β will form a plate in phase α.



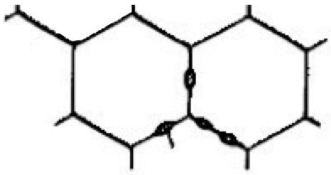
- Considering the case when interphase boundaries meet:

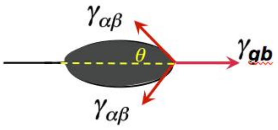
--- In the case that β forms at α grain boundaries:

The β particle shape depends on the interfacial energies. β meets the α grain boundary at an angle of θ, where

$$2\gamma_{\alpha\beta}\cos\theta = \gamma_{gb}$$

This is a balance of surface tensions.





--- Special case:

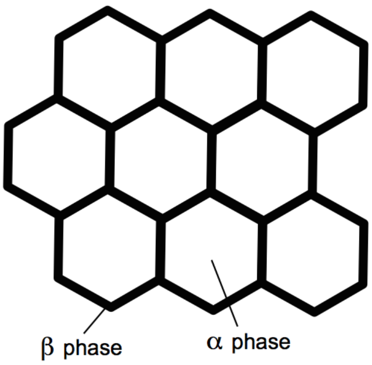
If $\gamma_{\alpha\beta} \le \frac{1}{2}\gamma_{gb}$

Then $\cos\theta = 1$

So $\theta = 0$

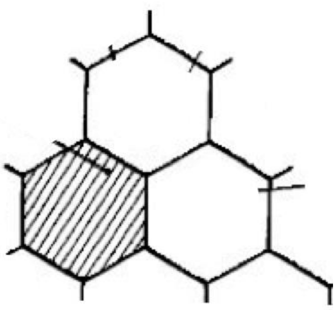
$$2\gamma_{\alpha\beta}\cos\theta = \gamma_{gb}$$

β phase will spread along all α grain boundaries causing interesting microstructure and material properties (such as Gallium Induced Structural Failure of an Aluminum Sheet).

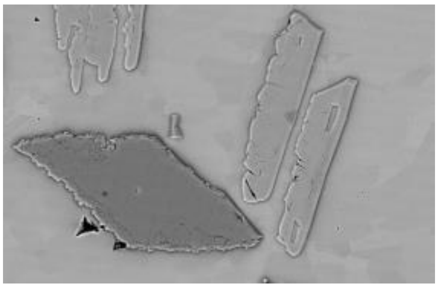


--- In some cases β can form their own grains adjacent to the α grains.

- If $\gamma_{\alpha\beta} = \gamma_{gb}$, the β grain shape will be also tetrakaidecahedral.



- Usually $\gamma_{\alpha\beta} \neq \gamma_{gb}$ and then complex β grain shapes occur. The β grain takes a shape that minimise the interfacial energy.



7. Importance of processing route

Metallic microstructures are influences by alloy constitution and geometry of the microstructure. Alloy constitution includes factors such as the number of phase present, the volume fraction of each phases and the composition of each phase and those factors are depending on their overall composition and temperature. The geometry of the microstructure are influenced by the size and spacing of each phase and the shape of each phases, which are also effected by processing route. Processing route plays an important role to the material microstructure and therefore their properties.